

Education

University of Chicago Mathematics	MS 1999, PhD 2005
Harvard University Mathematics and Women's Studies	BA 1998

Appointments

University of Chicago Professor of Data Science, Computer Science, and the College <i>Faculty Director</i> Data and Democracy Research Initiative	2025— 2025—
Cornell University Professor of Mathematics and Public Policy – <i>on leave</i> 2025–2026	2024–2025
Santa Fe Institute External Faculty	2024—
Tufts University John Dibiaggio Professor of Citizenship and Public Service Professor of Mathematics (previously Associate Professor, Assistant Professor) <i>Senior Fellow</i> Jonathan M. Tisch College of Civic Life <i>Director</i> Program in Science, Technology, & Society	2023–2024 2011–2024 2017–2025 2015–2021
University of Michigan Assistant Professor (postdoctoral)	2008–2011
University of California, Davis NSF VIGRE Postdoctoral Fellow	2005–2008

Research Interests

Data science for democracy, computational social choice theory, geometry and redistricting.
Science, technology, and society, computing and law, science policy, census data, privacy.
Random walks and Markov chains, partition problems, networks, algorithmic fairness.
Geometric group theory, geometric topology, hyperbolicity, metric geometry.

Selected Awards & Distinctions

Sloan Professor , MSRI Program in Algorithms, Fairness, and Equity	Fall 2023
Seelye Fellow , University of Auckland Department of Mathematics	2023
Radcliffe Fellow - Evelyn Green Davis Fellowship	2018–2019
Guggenheim Fellow	2018
Fellow of the American Mathematical Society	elected 2017
NSF C-ACCEL (PI) - Harnessing the Data Revolution: Network science of Census data	2019–2020
NSF grants (PI) - CAREER grant and continuous grants from Topology/Geometric Analysis	2009–2024
Professor of the Year , Tufts Math Society	2012–2013
AAUW Dissertation Fellowship	2004–2005
NSF Graduate Fellowship	1998–2002
Robert Fletcher Rogers Prize (Harvard Mathematics)	1995–1996

Political Geometry: Rethinking Redistricting in the U.S. with Math, Law, and Everything In Between

25 chapters, 475 pages. Birkhäuser Books 2022. [Preprint online](#). (eds. Moon Duchin, Olivia Walch)

see: [Introduction](#), [Compactness](#), [Communities of Interest](#), [Clustering](#), [Random Walks](#), [Ranked Choice Voting](#).

Grading Arrow's axioms

In submission. [Preprint](#). (with Suvadip Sana, Peter Rock, and Martin Wells)

How Banzhaf makes a Victor: Curing a distortion paradox in weighted voting

In submission. [Preprint](#). (with Jakob de Raij, Ariel Procaccia, and Jamie Tucker-Foltz)

Metric geometry for ranking-based voting: Tools for learning electoral structure

In submission. [Preprint](#). (with Kris Tapp)

The group perspective on fairness in multi-winner voting rules

In submission. [Preprint](#). (with Kevin Quinn)

Repetition effects in a sequential Monte Carlo sampler

In submission. [Preprint](#). (with Sarah Cannon and Daryl DeFord)

Proportionality for leaky coalitions

In submission. [Preprint](#). (with Gerdus Benadè, Chris Donnay, and Thomas Weighill)

Learning blocs and slates from ranked-choice ballots

In submission. [Preprint](#). (with Peter Rock, David Shmoys, Kris Tapp)

Mapper graphs for voting analysis

In preparation. [Preprint](#). (with Hazel Brenner, Emarie De La Nuez, and Jordan Phan)

Ranked choice voting and proportional representation

In preparation. [Preprint](#). (with Gerdus Benade, Ruth Buck, Dara Gold, and Thomas Weighill)

Spanning tree methods for redistricting: New methods for sampling and validation

SIAM Review, to appear. [Preprint](#). (with Sarah Cannon, Dana Randall, and Parker Rule)

VoteKit: A Python package for computational social choice research

Journal of Open Source Software **10** (109), 7477. [Open access](#).

(with Christopher Donnay, Jack Gibson, Zach Glaser, Andrew Hong, Malavika Mukundan, and Jennfier Wang)

Discrete geometry for electoral geography

Political Geography, Volume 109, March 2024. [Open access](#). (with Bridget Eileen Tenner)

Measuring segregation via analysis on graphs

SIAM Journal on Matrix Analysis and Applications (SIMAX), **44** (1) 2023, 80–105. [Preprint version](#).

(with James Murphy and Thomas Weighill)

Implementing partisan symmetry: Problems and paradoxes

Political Analysis **31** (3) 2023, 305–324. [Open access](#).

(with Daryl DeFord, Natasha Dhamankar, Mackenzie McPike, Gabe Schoenbach, and Ki-Wan Sim)

Redistricting for proportionality

The Forum: A Journal of Applied Research in Contemporary Politics, Vol. 20, No. 3-4, Jan 2023, 371–393.

[Open access](#). (with Gabe Schoenbach)

Blind justice: Algorithms and neutrality in the case of redistricting

Proceedings of 2nd ACM Symposium on Computer Science and Law (CS&Law), Nov 2022, 101–108. [Open access](#).

(with Doug Spencer)

Aggregating community maps

ACM Conference on Advances in GIS (SIGSPATIAL), Nov 2022, 1–12. [Open access](#). (with Erin Chambers, Ranthony Edmonds, Parker Edwards, JN Matthews, Anthony Pizzimenti, Chanel Richardson, Parker Rule, and Ari Stern)

Private numbers in public policy: Census, differential privacy, and redistricting

Harvard Data Science Review, Spec. Iss. 2, June 2022. [Open access](#). (w Aloni Cohen, JN Matthews, Bhushan Suwal)

The (homological) persistence of gerrymandering

Foundations of Data Science, Vol 4, Issue 4 (2022): 581–622. [Preprint version](#). (w Tom Needham and Thomas Weighill)

Recombination: A family of Markov chains for redistricting

Harvard Data Science Review. Issue 3.1, Winter 2021. [Open access](#). (with Daryl DeFord and Justin Solomon)

Census TopDown: The impact of differential privacy on redistricting

2nd Symposium on Foundations of Responsible Computing (FORC 2021), 5:1–5:22. [Open access](#).
(with Aloni Cohen, JN Matthews, and Bhushan Suwal)

Computational Redistricting and the Voting Rights Act

Election Law Journal, Volume 20, Number 4 (2021), 407–441. [Open access](#).
(with Amariah Becker, Dara Gold, and Sam Hirsch)

Models, Race, and the Law

Yale Law Journal Forum, Vol. 130 (March 2021). [Open access](#). (with Doug Spencer)

Clustering propensity: Segregation in networks

[Preprint](#). (with Emilia Alvarez, Everett Meike, and Marshall Mueller; appendix by Tyler Piazza)

Mathematics of nested districts: The case of Alaska

Statistics and Public Policy. Vol 7, No 1 (2020), 39–51. [Preprint version](#).
(w/ Sophia Caldera, Daryl DeFord, Sam Gutekunst, & Cara Nix)

A computational approach to measuring vote elasticity and competitiveness

Statistics and Public Policy. Vol 7, No 1 (2020), 69–86. [Open access](#). (with Daryl DeFord and Justin Solomon)

Redistricting reform in Virginia: Districting criteria in context

Virginia Policy Review, Volume XII, Issue II, Spring 2019, 120–146. [Preprint version](#). (with Daryl DeFord)

Locating the representational baseline: Republicans in Massachusetts

Election Law Journal, Volume 18, Number 4, 2019, 388–401. [Open access](#).
(with Taissa Gladkova, Eugene Henninger-Voss, Ben Klingensmith, Heather Newman, and Hannah Wheelen)

Geometry v. Gerrymandering

The Best Writing on Mathematics 2019, ed. Mircea Pitici. Princeton University Press.
reprinted from Scientific American, November 2018, 48–53. [Magazine version](#).

Gerrymandering metrics: How to measure? What's the baseline?

Bulletin of the American Academy for Arts and Sciences, Vol. LXII, No. 2 (Winter 2018), 54–58. [Preprint version](#).

Rebooting the mathematics of gerrymandering: How can geometry track with our political values?

The Conversation (online magazine), October 2017. [Open access](#). (with Peter Levine)

A formula goes to court: Partisan gerrymandering and the efficiency gap

Notices of the American Mathematical Society **64** No. 9 (2017), 1020–1024. [Open access](#). (with Mira Bernstein)

International mobility and U.S. mathematics

Notices of the American Mathematical Society **64**, No. 7 (2017), 682–683. [Open access](#).

Pure Mathematics Publications & Preprints

[open access links included](#)

Models of random spanning trees

Random Structures & Algorithms, Volume 68, Issue 3 (May 2026). [Preprint](#).
(with Eric Babson, Annina Iseli, Pietro Poggi-Corradini, Dylan Thurston, and Jamie Tucker-Foltz)

Conjugation curvature for Cayley graphs

Journal of Topology and Analysis, Vol 14, Number 02 (2022), 439–459. [Preprint version](#).
(with Assaf Bar-Natan and Robert Kropholler)

You can hear the shape of a billiard table: Symbolic dynamics and rigidity for flat surfaces

Commentarii Mathematici Helvetici, Vol 96, Issue 3 (2021), 421–463. [Preprint version](#).
(with Viveka Erlandsson, Christopher Leininger, and Chandrika Sadanand)

Stars at infinity in Teichmüller space

Geometriae Dedicata, Volume 213, 531–545 (2021). (with Nate Fisher) [Preprint version](#).

The Heisenberg group is pan-rational

Advances in Mathematics **346** (2019), 219–263. [Open access](#). (with Michael Shapiro)

Random nilpotent groups I

International Mathematics Research Notices, Vol. 2018, Issue 7 (2018), 1921–1953. [Open access](#).
(with Matthew Cordes, Yen Duong, Meng-Che Ho, and Ayla Sánchez)

Hyperbolic groups

in *Office Hours with a Geometric Group Theorist*, eds. M. Clay, D. Margalit, Princeton U Press (2017), 177–203. [Offprint](#).

Counting in groups: Fine asymptotic geometry

Notices of the American Mathematical Society **63**, No. 8 (2016), 871–874. [Open access](#).

A sharper threshold for random groups at density one-half

Groups, Geometry, and Dynamics **10**, No. 3 (2016), 985–1005. [Open access](#).

(with Katarzyna Jankiewicz, Shelby Kilmer, Samuel Lelièvre, John M. Mackay, and Ayla Sánchez)

Equations in nilpotent groups

Proceedings of the AMS **143** (2015), 4723–4731. [Open access](#). (with Hao Liang and Michael Shapiro)

Statistical hyperbolicity in Teichmüller space

GAF, Volume 24, Issue 3 (2014), 748–795. [Preprint version](#). (with Howard Masur and Spencer Dowdall)

Fine asymptotic geometry of the Heisenberg group

Indiana University Mathematics Journal 63 No. 3 (2014), 885–916. [Preprint version](#). (with Christopher Mooney)

Pushing fillings in right-angled Artin groups

JLMS, Vol 87, Issue 3 (2013), 663–688. [Preprint version](#). (w Aaron Abrams, Noel Brady, Pallavi Dani, Robert Young)

Spheres in the curve complex

Ahlfors–Bers VI, Contemp. Math. **590** (2013), 1–8. [Preprint version](#). (with Howard Masur and Spencer Dowdall)

The sprawl conjecture for convex bodies

Experimental Mathematics, Volume 22, Issue 2 (2013), 113–122. [Offprint](#). (w Samuel Lelièvre, Christopher Mooney)

Filling loops at infinity in the mapping class group

Michigan Math. J. **61** (4) 2012, 867–874. [Preprint version](#). (w Aaron Abrams, Noel Brady, Pallavi Dani, Robert Young)

The geometry of spheres in free abelian groups

Geom. Dedicata, Volume 161, Issue 1 (2012), 169–187. [Preprint version](#). (with Samuel Lelièvre and Christopher Mooney)

Statistical hyperbolicity in groups

Algebraic and Geometric Topology **12** (2012) 1–18. [Open access](#). (with Samuel Lelièvre and Christopher Mooney)

Length spectra and degeneration of flat metrics

Inventiones Math., Volume 182, Issue 2 (2010), 231–277. [Preprint version](#). (w Christopher Leininger, Kasra Rafi)

Divergence of geodesics in Teichmüller space and the mapping class group

Geometric and Functional Analysis, Volume 19, Issue 3 (2009), 722–742. [Preprint version](#). (with Kasra Rafi)

Curvature, stretchiness, and dynamics

In the Tradition of Ahlfors and Bers IV, Contemp. Math. **432** (2007), 19–30. [Offprint](#).

Geodesics track random walks in Teichmüller space

PhD Dissertation, University of Chicago 2005.

Teaching

Courses Developed or Customized**Data and the State: How Governments "See" People and Places**

Interdisciplinary undergraduate course teaching Python-based geospatial tools with policy applications.

Modeling Democracy

Interdisciplinary PhD course on computational social choice and applied modeling of democratic systems.

Mathematics of Social Choice | sites.tufts.edu/socialchoice

Voting theory, impossibility theorems, redistricting, theory of representative democracy, metrics of fairness.

Have designed and taught variants at entry level and math-major level.

History of Mathematics | sites.tufts.edu/histmath

Social history of mathematics, organized around episodes from antiquity to present. Themes include materials and technologies of creation and dissemination, axioms, authority, credibility, and professionalization. In-depth treatment of mathematical content from numeration to cardinal arithmetic to Galois theory.

Reading Lab: Mathematical Models in Social Context | sites.tufts.edu/models

One hr/wk discussion seminar of short but close reading on topics in mathematical modeling, including history of psychometrics; algorithmic bias; philosophy of statistics; problems of model explanation and interpretation.

Reading Lab: Classification | sites.tufts.edu/classification

Discussion-based seminar of close reading on topics in classifications and taxonomies, including censuses; race and ethnicity; academic disciplines, definition in math and law; chemical elements; model organisms; sex and gender.

Geometric Literacy

Module-based graduate topics course. Modules have included: p -adic numbers, hyperbolic geometry, nilpotent geometry, Lie groups, convex geometry and analysis, the complex of curves, ergodic theory, the Gauss circle problem.

Randomized Algorithms (graduate/undergraduate topics course)

Markov Chains (graduate topics course)

Teichmüller Theory (graduate topics course)

Fuchsian Groups (graduate topics course)

Continued Fractions and Geometric Coding (undergraduate topics course)

Mathematics for Elementary School Teachers (inquiry-based course for pre-service teachers)

Standard Courses

Mathematical Modeling and Computation (with Python), Discrete Mathematics, Calculus I-II-III, Intro to Proofs, Linear Algebra, Complex Analysis, Differential Geometry, Abstract Algebra, Graduate Real Analysis

Selected Talks and Lectures

Keynote Lecture

International School & Conference on Network Science (NetSci), Boston, MA

June 2026

Keynote Lecture

Computational Social Choice Workshop (COMSOC), Vienna, Austria

October 2025

Invited Talk

Autumn General Meeting of the American Philosophical Society, Philadelphia, PA

November 2024

Plenary Lecture

SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, GA

October 2024

Plenary Lecture

Symposium on Computational Geometry (SoCG), Dallas, TX

June 2023

Distinguished Plenary Lecture

75th Anniversary Meeting of Canadian Mathematical Society, Ottawa, Ontario

June 2021
online (COVID)

BMC/BAMC Public Lecture

Joint British Mathematics/Applied Mathematics Colloquium, Glasgow, Scotland

April 2021
online (COVID)

Gerald and Judith Porter Public Lecture

AMS-MAA-SIAM, Joint Mathematics Meetings, San Diego, CA

January 2018

Mathematical Association of America Distinguished Lecture

MAA Carriage House, Washington, DC

October 2016

American Mathematical Society Invited Address

AMS Eastern Sectional Meeting, Brunswick, ME

September 2016

Named University Lectures

- University Lecture, Data Science Distinguished Lecture Cornell University	February 2024
- Martha Davenport Heard Lecture Wellesley College	February 2024
- 47 Lecture Pomona College	October 2023
- Seelye Public Lecture University of Auckland, New Zealand	March 2023
- Lorne Campbell Lecture Queen's University, Ontario	December 2022
- Plancherel Lecture Université de Fribourg, Switzerland	October 2022
- Loeb Lectures in Mathematics Washington University in St. Louis	April 2022
- Mathematics and Natural Sciences Divisional Lecture Reed College	March 2022
- Parsons Lecture UNC Asheville	October 2020
- Math, Stats, CS, and Society Macalester College	October 2019
- MRC Public Lecture Stanford University	May 2019
- Freedman Memorial Colloquium Boston University	March 2019
- Julian Clancy Frazier Colloquium Lecture U.S. Naval Academy	January 2019
- Barnett Lecture University of Cincinnati	October 2018
- School of Science Colloquium Series The College of New Jersey	March 2018
- Kieval Lecture Cornell University	February 2018

Mathematics and Computational Science

- David Blackwell Seminar in Statistics and Data Science University of Washington	October 2025
- Statistics and Data Science Department Colloquium Northwestern University	October 2025
- Online Social Choice and Welfare Seminar	August 2023
- Data Matters Public Lecture Data Science Institute, Brown University	April 2023
- Computational Social Choice Seminar Center for Mathematical Social Science, Auckland	March 2023
- Mathematics Department Colloquium Northeastern University	February 2023
- Societal Considerations and Applications Simons Institute for the Theory of Computing	November 2022
- ACM Symposium on Computer Science and Law Washington, DC	November 2022
- Modeling democracy (three hours) Modern Math Workshop, Puerto Rico	October 2022
- Mathematics Department Colloquium University of Michigan	September 2022
- Mathematics Department Colloquium UC Berkeley	April 2022
- Mechanism Design for Social Good (MD4SG) Colloquium MD4SG Initiative	November 2020
- Integer programming and combinatorial optimization (two talks) Georgia Tech	May 2021
- Data Science for Social Good (DS4SG) Workshop Georgia Tech	November 2020
- Computer Science Department Colloquium Georgetown University	September 2020
- Complex Systems Colloquium Santa Fe Institute	July 2020
- Women in Data Science Conference Microsoft Research New England	March 2020
- Societal Concerns in Algorithms and Data Analysis Weizmann Institute	December 2018

Law and STS (Science, Technology, and Society)

- STS Colloquium Cornell University	March 2024
- Voting Rights Panel Rothgerber Conference, University of Colorado Law School	April 2023
- Math, Law, and Racial Fairness Justice Speaker Series, University of South Carolina	November 2020
- Voting Rights Conference Northeastern Public Interest Law Program	September 2020
- The Mathematics of Accountability Sawyer Seminar, Anthropology, Johns Hopkins	February 2020
- Program in Public Law Panel Duke Law School	October 2019
- STS Circle Harvard Kennedy School of Government	September 2019
- Data, Classification, and Everyday Life Symposium Rutgers Center for Cultural Analysis	January 2019
- Science Studies Colloquium UC San Diego	January 2019
- Arthur Miller Lecture on Science and Ethics MIT Program in Science, Tech, and Society	November 2018
- Cyber Security, Law, and Society Alliance Boston University	September 2018
- Tech/Law Colloquium Series Cornell Tech	November 2017

Social Science and Public Policy

- Sociology Department Colloquium University of Chicago	April 2026
- Masterclass Harris School, University of Chicago	April 2026
- Data and Democracy Scholar Talk Harris School, University of Chicago	April 2023
- Censuses and Racial Classification COMPASS, University of Auckland	March 2023
- Econ/CS Seminar Harvard	October 2022
- Can Algorithms Bend the Arc Towards Fairness? Algorithmic Justice Project, UNM/SFI	March 2022
- Data Linkage Seminar Massive Data Institute, McCourt School of Public Policy	August 2021
- The Long 19th Amendment: Women, Voting, and American Democracy Radcliffe Institute	Nov–Dec 2020
- "The New Math" for Civil Rights Social Justice Speaker Series, Davidson College	November 2020
- Quantitative Research Methods Workshop Yale Center for the Study of American Politics	February 2020
- Political Analysis Workshop Indiana University	November 2019
- Redistricting 2021 Seminar University of Chicago Institute of Politics	May 2019
- Geography of Redistricting Conference Keynote Harvard Center for Geographic Analysis	May 2019
- Political Analytics Conference Harvard University	November 2018
- Quantitative Collaborative University of Virginia	March 2018
- Clough Center for the Study of Constitutional Democracy Boston College	November 2017
- Data for Black Lives Conference MIT	November 2017
- Constitution Day Lecture Rockefeller Center for Public Policy, Dartmouth College	September 2017
- Quantitative Social Science Dartmouth College	September 2017

Visiting Lectures

- Election Law Harvard Law School	Spring 2022, Spring 2026
- Law and Algorithms Boston University	Spring 2024
- Normative, Legal, and Empirical Analyses of Discrimination Yale Law School	Spring 2024
- Optimized Democracy Harvard (CS)	2021, 2022, 2023, 2024, 2025
- Law of Democracy Stanford Law School	Fall 2022
- A Democracy Initiative Harvard Law School	Fall 2022
- Election Law Yale Law School	Spring 2022
- Privacy, Policy, and the U.S. Census University of Chicago (CS)	Spring 2022

Program Development

Director, PI of **Data and Democracy Lab** (formerly MGGG Redistricting Lab) data-democracy.org

Multidisciplinary research lab with postdocs, research staff, and undergraduate researchers drawn from mathematics, computer science, geography, policy. Hosts law student externs. Provided public mapping support for roughly 140 localities after 2020 Census data released.

Support includes NSF, Crankstart Foundation, Democracy Fund, Kelson Foundation, Sloan Foundation, Thornburg Foundation, Arnold Foundation.

Director of **Structural Democracy Fellowship Program**

Fellowship program distributing research funding to cohort of 17 U.S. and international fellows. Tied to weekly online seminar and special issue of Harvard Data Science Review, titled *Designing Democracy*.

Funded by gift from Crankstart Foundation.

Co-Founder, Program Director of **Science, Technology, and Society Program** sts.tufts.edu

Interdisciplinary program offering a major and minor, with ~40 affiliated faculty. Initiated popular weekly lunch seminar and developed Reading Lab courses on topics from Automation to Classification to Life to Energy.

Program Organization

Semester Program in *Algorithms, Fairness, and Equity*, Fall 2023
Mathematical Sciences Research Institute, Berkeley CA

Program hosted ~50 research members on topics connected to mechanism design, fair partitioning, and fair ML.

Short workshops and training programs

- Ranked Choice Modeling Session 2025 (2 days, planned for June 2025 at Cornell)
- Ranked Choice Modeling Session 2023 (1 day, 20 participants from think tanks and NGOs)
- GeoData Bootcamp 2020 (2 weeks, 20 students from around the country)
- Mapping Training 2020 (1 week, 30 students from around the country)
- Graphs and Networks Workshop 2020 (1 day, 500 live participants)
- Data for Election Administration 2019, 2021 (multi-day, dozens of administrators and scholars)

Program Building Research and mentorship programs

- Voting Rights Data Institute 2018, 2019, 2023, 2025
Six-week summer research programs hosting up to 52 undergraduate and graduate students, respectively, with dozens of visitors from math, CS, law, political science, geography, urban planning, and more.
- Polygonal Billiards Research Cluster 2017, Random Groups Research Cluster 2014
Five-week intensive summer research programs for vertically integrated groups of 12-14 undergraduate, graduate, postdoctoral, and junior faculty researchers, combining experimental and theoretical work.
- Directed Reading Program and DRP Network sites.google.com/view/drp-network/
Co-founded highly successful near-peer mentoring program in 2003 at UChicago. Now exists at >40 math departments as grad-student-run reading program with excellent outcomes for broadening participation in mathematics. Secured NSF grant to expand the program to more campuses and to fund social science research on outcomes.

Graduate Advising

Nate Fisher (PhD 2021), Sunrose Shrestha (PhD 2020), Ayla Sánchez (PhD 2017),
Kevin Buckles (PhD 2015), Mai Mansouri (MS 2014)

Co-advisor or outside committee member

Phousawanh Peaungvongpakdy (Washington State), Chris Donnay (PhD 2024 Ohio State),
Chris Coscia (PhD 2020 Dartmouth College)

Postdoctoral Advising

Principal supervisor Hane Lee (2025–), Shubham Singh (2025–2026), Amariah Becker (2020–2021), Thomas Weighill (2019–2020)

Co-supervisor Sam Zhang (SFI 2024–2025), Daryl DeFord (MIT 2018–2020), Rob Kropholler (2017–2020), Hao Liang (2013–2016)

Selected Professional Service and Public-Facing Work

Program committees and editorial boards

Harvard Data Science Review	since 2019
ACM Conference on Fairness, Accountability, and Computing (FAccT)	2022
Symposium on Foundations of Responsible Computing (FORC)	2021
Advances in Mathematics	2018–2023

Committee on Science Policy

American Mathematical Society	2020–2022
-------------------------------	-----------

Amicus Brief of Mathematicians, Law Professors, and Students

<i>principal co-authors: Guy-Uriel Charles and Moon Duchin</i>	2019
Supreme Court of the United States, in <i>Rucho v. Common Cause</i> - cited in dissent	

Expert work for redistricting litigation

<i>reports, deposition, and/or trial testimony</i>	2018—
Pennsylvania, Wisconsin, North Carolina, Alabama, South Carolina, Texas, Boston, Georgia	

League of Women Voters of Pa. v. Commonwealth, 178 A.3d 737 (Pa. 2018); *Johnson v. Wis. Elections Comm’n*, No. 2021AP1450-OA, 2022 WL 621082 (Wis. Mar. 3, 2022); *N.C. League of Conservation Voters v. Hall*, Nos. 21 CVS 015426, 21 CVS 500085, 2022 WL 2610499 (N.C. Super. Ct. Feb. 23, 2022); *Allen v. Milligan*, No. 2:21-cv-01530-AMM (N.D. Ala.) (three-judge ct.); *Carter v. Chapman*, No. 7 MM 2022, 2022 WL 702894 (Pa. Mar. 9, 2022); *S.C. State Conf. of the NAACP v. Alexander*, No. 3:21-cv-03302-MBS-TJH-RMG (D.S.C.) (three-judge ct.); *LULAC v. Abbott*, No. 3:21-CV-00259-DCG-JES-JVB (W.D. Tex.) (three-judge ct.); *O’Shea v. Boston City Council*, No. 1:22-cv-12048-PBS (D. Mass.); *Ga. State Conf. of the NAACP v. Georgia*, No. 1:21-cv-05338-ELB-SCJ-SDG (N.D. Ga.) (three-judge ct.); *Alpha Phi Alpha Fraternity, Inc. v. Raffensperger*, No. 1:21-cv-05337-SCJ (N.D. Ga.)

Presenter on Public Mapping, Statistical Modeling

National Conference of State Legislatures	2019, 2020
---	------------

Committee on The Future of Voting: Accessible, Reliable, Verifiable Technology

National Academies of Science, Engineering, and Medicine	2017–2018
--	-----------

Committee on the Human Rights of Mathematicians

American Mathematical Society	2016–2019
-------------------------------	-----------